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Multiclass Classification using Logistic Regression :-

Assignment :-

Predict Fruit Type (Apple, Orange, Banana) based on:

Weight

Sweetness

Size

Step 1: Import Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report
```

Step 2: Create a Simple Multiclass Dataset

Target Variable: Classes:

Apple → Apple

Orange → Orange

Banana → Banana

```
data = {
    'Weight_g': [
        150, 160, 170, 145, 180, 155, 165, 175, 140, 185,
        200, 210, 190, 220, 205, 215, 195, 225, 230, 185,
        110, 120, 125, 115, 130, 105, 135, 118, 128, 122
    ],

    'Sweetness': [
        6.2, 6.5, 6.8, 5.9, 7.0, 6.3, 6.7, 6.9, 5.8, 7.1,
        7.5, 7.8, 7.2, 8.0, 7.6, 7.9, 7.3, 8.2, 8.4, 7.0,
        8.5, 8.8, 9.0, 8.6, 9.2, 8.3, 9.4, 8.7, 9.1, 8.9
    ],

    'Size_cm': [
        7.0, 7.2, 7.5, 6.8, 7.8, 7.1, 7.4, 7.6, 6.7, 7.9,
        8.0, 8.2, 7.8, 8.5, 8.1, 8.4, 7.9, 8.6, 8.8, 7.7,
        12.0, 13.0, 13.5, 12.5, 14.0, 11.8, 14.5, 12.8, 13.8, 13.2
    ],

    'Fruit': [
        'Apple', 'Apple', 'Apple', 'Apple', 'Apple', 'Apple', 'Apple', 'Apple',
        'Orange', 'Orange', 'Orange', 'Orange', 'Orange', 'Orange', 'Orange', '
        'Banana', 'Banana', 'Banana', 'Banana', 'Banana', 'Banana', 'Banana', '
    ]
}

df = pd.DataFrame(data)
df
```

Step 3: Explore the Dataset

```
df.shape
```

```
(30, 4)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype  
---  ------     --------------  -----  
 0   Weight_g   30 non-null     int64  
 1   Sweetness  30 non-null     float64
 2   Size_cm    30 non-null     float64
 3   Fruit      30 non-null     object 
dtypes: float64(2), int64(1), object(1)
memory usage: 1.1+ KB
```

```
df.describe()
```

Step 4: Split Features and Target

```
X = df[['Weight_g', 'Sweetness', 'Size_cm']]
y = df['Fruit']
```

Step 5: Train-Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Step 6: Create and Train Logistic Regression Model

```
model = LogisticRegression(multi_class='multinomial', solver='lbfgs', max_iter=
model.fit(X_train, y_train)
```

Step 7: Model Prediction and Evaluation

```
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))
```

```
Accuracy: 0.8333333333333334
Classification Report:
               precision    recall  f1-score   support

       Apple       1.00      0.50      0.67         2
      Banana       1.00      1.00      1.00         2
      Orange       0.67      1.00      0.80         2

    accuracy                           0.83         6
   macro avg       0.89      0.83      0.82         6
weighted avg       0.89      0.83      0.82         6
```

Step 8: Visualization

```
fruit_codes = df['Fruit'].map({'Apple': 0, 'Orange': 1, 'Banana': 2})

plt.scatter(df['Weight_g'], df['Sweetness'], c=fruit_codes, cmap='viridis')
plt.xlabel('Weight (g)')
plt.ylabel('Sweetness')
plt.title('Multiclass Data Distribution')
plt.colorbar(label='Fruit Class')
plt.show()
```

Step 9: Prediction Using User Input

```
Weight_g = float(input("Enter Weight in grams: "))
Sweetness = float(input("Enter Sweetness: "))
Size_cm = float(input("Enter Size in cm: "))

new_data = [[Weight_g, Sweetness, Size_cm]]
prediction = model.predict(new_data)

print("Predicted Fruit Type:", prediction[0])
```

```
Enter Weight in grams: 50
Enter Sweetness: 4.9
Enter Size in cm: 5.7
Predicted Fruit Type: Banana
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWa
  warnings.warn(
```

```
import pickle
```

```
pickle.dump(model,open('MCModel.pkl','wb'))
```

Flask Code

Save this file with 'app.py'

```
from flask import Flask, render_template, request
import pickle

app = Flask(__name__)

# Load Trained Model
model = pickle.load(open("MCModel.pkl", "rb"))


@app.route('/')
def home():
```

```
    return render_template("index.html")


@app.route('/predict', methods=['POST'])
def predict():

    # Read input values
    weight_g = float(request.form['weight_g'])
    sweetness = float(request.form['sweetness'])
    size_cm = float(request.form['size_cm'])

    # Prediction
    prediction = model.predict([[weight_g, sweetness, size_cm]])

    # Convert class into fruit type
    fruit = prediction[0]

    return render_template(
        "index.html",
        prediction_text=f"Predicted Fruit Type : {fruit}"
    )


if __name__ == "__main__":
    app.run(debug=True)
```

```
 * Serving Flask app '__main__'
 * Debug mode: on
INFO:werkzeug:WARNING: This is a development server. Do not use it in a producti
 * Running on http://127.0.0.1:5000
INFO:werkzeug:Press CTRL+C to quit
INFO:werkzeug: * Restarting with watchdog (inotify)
```

HTML Code

Save this file with 'index.html'

```
<!DOCTYPE html>
<html lang="en">

<head>

<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Fruit Type Predictor</title>

<style>
```

```
*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:Arial, Helvetica, sans-serif;
}

body{
    background:#eef5ff;
    display:flex;
    justify-content:center;
    align-items:center;
    min-height:100vh;
}

.container{
    width:480px;
    background:white;
    padding:35px;
    border-radius:15px;
    box-shadow:0px 10px 25px rgba(0,0,0,.2);
}

h1{
    text-align:center;
    color:#1565c0;
}

h3{
    text-align:center;
    color:#555;
    margin-top:8px;
    margin-bottom:25px;
}

label{
    font-weight:bold;
    display:block;
    margin-top:15px;
    margin-bottom:6px;
}

input{
    width:100%;
    padding:12px;
    border:1px solid #ccc;
    border-radius:8px;
    font-size:16px;
}

button{
```

```
        width:100%;
        margin-top:25px;
        padding:13px;
        background:#1565c0;
        color:white;
        border:none;
        border-radius:8px;
        font-size:18px;
        cursor:pointer;
        transition:.3s;
    }

    button:hover{
        background:#0d47a1;
    }

    .result{
        margin-top:25px;
        background:#e8f5e9;
        padding:18px;
        border-radius:8px;
        text-align:center;
        color:green;
        font-size:21px;
        font-weight:bold;
    }

    footer{
        text-align:center;
        margin-top:20px;
        color:#777;
        font-size:13px;
    }

    </style>

    </head>

    <body>

    <div class="container">

    <h1>Fruit Type Predictor</h1>

    <h3>Multiclass Classification using Logistic Regression</h3>

    <form action="/predict" method="POST">

    <label>Weight (g)</label>

    <input
```

```
type="number"
name="weight_g"
step="0.1"
min="1"
placeholder="Enter Weight"
required>

<label>Sweetness</label>

<input
type="number"
name="sweetness"
step="0.1"
min="0"
placeholder="Enter Sweetness"
required>

<label>Size (cm)</label>

<input
type="number"
name="size_cm"
step="0.1"
min="1"
placeholder="Enter Size"
required>

<button type="submit">
Predict Fruit
</button>

</form>

{% if prediction_text %}

<div class="result">
{{ prediction_text }}
</div>

{% endif %}

<footer>
Fruit Type Prediction
</footer>

</div>

</body>

</html>
```